

Topics to discuss

- What is Integration
- What is numerical Integration
- Difference between Integration & numerical Integration
- Why to use numerical Integration
- Method to solve numerical Integration problem.

What is Integration ?

→ Integration is a mathematical process used to find the total accumulation of a quantity, most commonly the area under a curve.

Indefinite Integral

→ gives a function + constant C

→ no limits

$$\rightarrow \int dx = x + C$$

Definite Integral

→ gives a number

→ limits a to b

$$\rightarrow \int_a^b dx = b - a$$

Example :

Equation of straight line
passes through $(1,2)$ & $(4,2)$

will be :

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

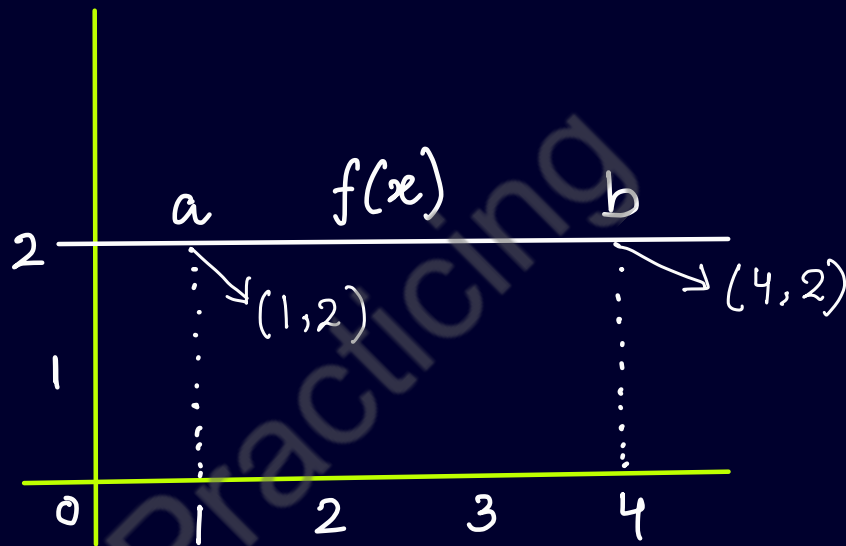
$$y - 2 = 0$$

$$y = 2$$

$$\therefore y = f(x) = 2 \rightarrow \text{st. line}$$

Now lets use integration to find area under the curve, $y=2$

$$\Rightarrow \int_a^b f(x) dx = \int_1^4 2 dx = 2 \int_1^4 dx = 2 [x]_1^4 = 2 \times (4-1) = 6$$



What is numerical Integration ?

→ In numerical Integration, we do not try to find an exact formula. Instead, we approximate the area using numbers by dividing the region into small parts (rectangles, trapezoids, etc)

This is used when the function is too complicated or unknown. The result is an approximation, but it can be made very accurate.

$$\int_a^b f(x) dx = \int_a^b y dx$$

$$h = \frac{b-a}{n}$$

$$x_0 = a$$

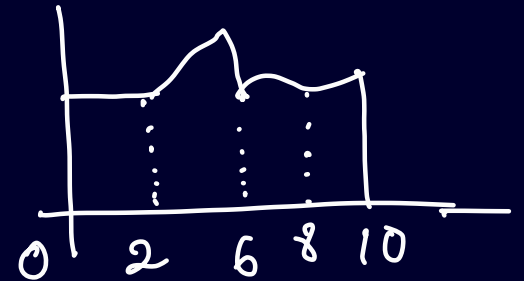
$$x_1 = x_0 + h$$

$$x_2 = x_0 + 2h$$

$$x_3 = x_0 + 3h$$

$$\vdots$$

$$x_n = x_0 + nh = b$$



Difference between Integration & Numerical Integration

Analytical Integration	Numerical Integration
• Exact value using formulas • Need known function	• Approximate value using calculations • Works even with complex or tabulated data

Why to use Numerical Integration ?

- The function has no closed-form integral.
- The integral is too complex to solve analytically.
- Data is given in tabular or experimental form.
- Computer-based approximation is required.
- Practical engineering or scientific problems need quick estimates.

Methods to solve Numerical Integration Problems

The main family of methods used to solve numerical integration problems is based on the Newton-Cotes formulas. These formulas approximate the integral by using polynomial interpolation over equally spaced points. Under Newton-Cotes, we have several well-known methods:

- Trapezoidal Rule
- Simpson's $\frac{1}{3}$ Rule
- Simpson's $\frac{3}{8}$ Rule
- Boole's Rule
- Weddle's Rule

Newton-Cotes formulas :

The formula is :

$$\int_{x_0}^{x_n} y dx = nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 + \dots \right]$$

This is known as general newton's cotes formula.

1) Trapezoidal Rule · (two point formula)

- Approximate area under curve with trapezoids.

$$\boxed{n=1}$$

The formula is :

$$\int_{x_0}^{x_n} y dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n]$$

This is known as Trapezoidal rule.

2) Simpson's $\frac{1}{3}$ Rule : (three point formula)

- Uses quadratic (parabolic) curves to approximate the sections.

$$\boxed{n=2}$$

The formula is:

$$\int_{x_0}^{x_n} y \, dx = \frac{h}{3} \left[y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n \right]$$

This is known as Simpson's $\frac{1}{3}$ Rule.

3) Simpson's $\frac{3}{8}$ Rule : (four point formula)

- Uses cubic polynomials for approximation.

$$\boxed{n=3}$$

The formula is :

$$\int_{x_0}^{x_n} y dx = \frac{3h}{8} \left(y_0 + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-2} + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3}) + y_n \right)$$

This is known as Simpson's $\frac{3}{8}$ Rule.

4) Boole's Rule : (five point formula)

- A higher-order formula that uses fourth degree polynomial. $\boxed{n=4}$

$$\int_{x_0}^{x_n} y dx = \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 + 32y_5 + 12y_6 + 32y_7 + 14y_8 + \dots)$$

where, n is the multiple of 4.

This is known as boole's rule.

5) Weddle's Rule : (seven point formula)

- Uses sixth-degree polynomial to approximate the integral.

$$\boxed{n=6}$$

$$\int_{x_0}^{x_n} y \, dx = \frac{3h}{10} \left[(y_0 + 5y_1) + (y_2 + 6y_3) + (y_4 + 5y_5) + y_6 + \right. \\ (y_6 + 5y_7) + (y_8 + 6y_9) + (y_{10} + 5y_{11}) + y_{12} + \\ (y_{12} + 5y_{13}) + (y_{14} + 6y_{15}) + (y_{16} + 5y_{17}) + y_{18} \\ \left. + \dots \right]$$

* n must be multiple of 6.

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